

Advanced (Habanero)

Q1. What is the expanded form of $(x + 3)^2$?

- (A) $x^2 + 6x + 9$
(B) $x^2 + 3x + 9$
(C) $x^2 - 6x + 9$
(D) $x^2 - 3x - 9$
(E) $x^2 + 9$

Q2. Which of the following is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
(B) $x^2 - 4x - 4$
(C) $x^2 + 2x - 4$
(D) $x^2 - 2x + 4$
(E) $x^2 - 4$

Q3. What is the factored form of $x^2 + 10x + 25$?

- (A) $(x + 5)^2$
(B) $(x - 5)^2$
(C) $(x + 2)(x + 8)$
(D) $(x - 2)(x - 8)$
(E) $(x + 1)(x + 9)$

Q4. What is the expanded form of $(x - 2)^2$?

- (A) $x^2 + 4x + 4$
(B) $x^2 - 4x + 4$
(C) $x^2 - 2x - 4$
(D) $x^2 - 4x - 4$
(E) $x^2 + 4$

Q5. Which of the following is NOT a perfect square trinomial?

- (A) $x^2 + 6x + 9$
(B) $x^2 + 2x + 1$
(C) $x^2 - 4x + 4$
(D) $x^2 + 2x - 1$
(E) $x^2 - 2x - 1$

Q6. What is the factored form of $x^2 - 14x + 49$?

- (A) $(x + 7)^2$
(B) $(x - 7)^2$
(C) $(x + 1)(x - 13)$
(D) $(x - 1)(x + 13)$
(E) $(x + 2)(x - 12)$

Q7. What is the expanded form of $(2x + 1)^2$?

- (A) $4x^2 + 4x + 1$
(B) $4x^2 + 2x + 1$
(C) $4x^2 - 2x + 1$
(D) $4x^2 - 4x + 1$
(E) $4x^2 + 1$

Q8. Which of the following is a perfect square trinomial?

- (A) $x^2 + 2x - 4$
(B) $x^2 - 2x - 4$
(C) $x^2 + 2x + 1$
(D) $x^2 - 2x + 1$
(E) $x^2 - 4$

Open Ended Questions (FRQ)**Q9.** Factor the expression $x^2 + 12x + 36$.**Q10.** Expand the expression $(3x - 2)^2$ and simplify.

Chef's Tips

- Read each question carefully.
- Show all work for free response questions.
- Use the back of the page if necessary.